

Name:

Roll No.

Batch:

Signature :

MS101 – Makerspace
2023-24/I (Autumn Semester)

Aug 23, 2023 (Wed)

EE Quiz - 1

Time: 45 min

Marks: 35

1. This **Question-cum-Answer Booklet** has 4 pages and 12 questions.
2. Write your **answers only in the space provided for answers**. Answers written at any other place will not be checked. You may use the page margins for rough work.
3. No explanations / clarifications will be given to any of the questions.
4. No negative marks for wrong answers, however steps are required for all numerical answers.

1. State whether the following statement is TRUE or FALSE. Justify your answer in just *one* sentence.

A step-up transformer can be considered as an amplifier since its secondary voltage is larger than the primary voltage.

Marks: 2

Answer: TRUE / FALSE

One-sentence justification: -----

2. A practical voltage source has an open circuit voltage of 10 V. When a $1.75\text{ k}\Omega$ resistor is connected as load across its terminals, the terminal voltage drops to 8.75 V. What is the source resistance in ohms?
No marks without steps.

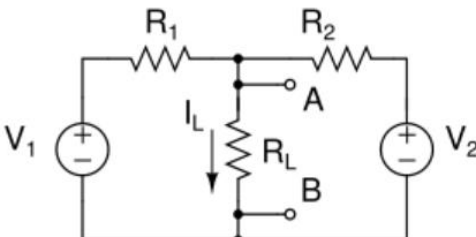
Marks: 3

Answer: ----- Ω

3. A practical current source is modelled as an ideal current source of 100 mA in parallel with a resistor of $240\text{ k}\Omega$. A load resistance of $10\text{ k}\Omega$ is connected across this source. What is the load current?
No marks without steps.

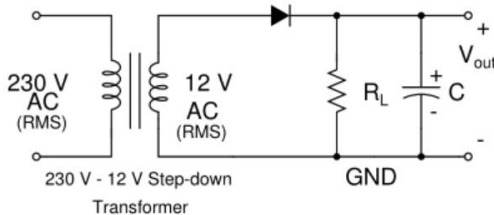
Marks: 2

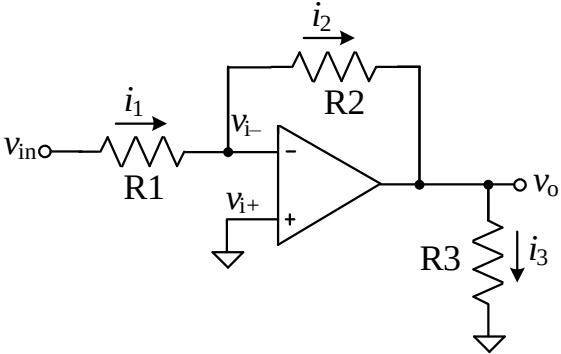
Answer: ----- mA

4. A resistive network is shown below. $R_1 = R_2 = R_L = 3000\ \Omega$. $V_1 = -15\text{ V}$, and $V_2 = 6\text{ V}$. Using superposition theorem, calculate V_{AB} and I_L. No marks without steps. 	Marks: 4 (= 3 + 1) Answers: $V_{AB} =$ V $I_L =$ mA
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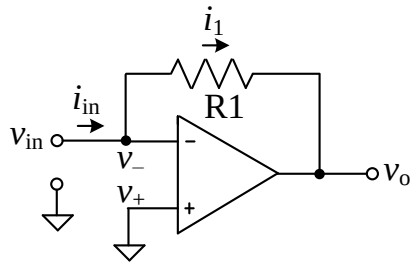
Show steps here:	
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5. The current through a <i>pn</i> junction diode in the forward region can be approximated as: $i_d = I_S \cdot \exp [v/V_T]$, where v is the voltage across the diode, I_S is the reverse-saturation current and V_T is the thermal voltage. If $V_T = 25\text{ mV}$, and $I_S = 10^{-13}\text{ A}$. Calculate i_d for $v = 0.6\text{ V}$. Your current should be precise to the first decimal place.	Marks: 2 Answer: $i_d =$ mA
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6. The circuit diagram of an unregulated DC power supply which employs a capacitive filter is shown below. The secondary voltage of the transformer is 12 Vrms. If resistor R_L is removed from the circuit, what will be V_{out} in volts? Justify your answer in <i>one</i> sentence. 	Marks: 3 (= 1 + 2) Answer:... V Justification (<i>one</i> sentence):
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7. How many conductors are needed for 8 single-ended signals?	Marks: 1 Answer:
8. A function generator having source resistance of $50\ \Omega$ is connected to an amplifier. Which of the following options gives the highest voltage at the amplifier's input? A) $R_{in} = 50\ \Omega$ B) $R_{in} \gg 50\ \Omega$ C) $R_{in} \ll 50\ \Omega$	Marks: 1 Answer: A B C
9. Mark <u>all true</u> statements regarding <u>virtual short</u> across the input terminals of an op amp. (A) Virtual short means zero input currents into the input terminals and zero voltage across them. (B) Virtual short is not applicable if the op amp output reaches its higher or lower saturation voltage. (C) Virtual short requires the positive and negative supply voltages to have the same magnitude. (D) Virtual short requires the non-inverting input to be grounded.	Marks: 2 All correct statements: A B C D
10. In the op-amp circuit shown in the figure, $R_1 = 100\ \text{k}\Omega$, $R_2 = 50\ \text{k}\Omega$, $R_3 = 100\ \Omega$. Find the voltage, current, and power gains. Is it serving as an amplifier? 	Marks: 4 (= 1 x 4) Answer: Voltage gain $A_v = v_o / v_{in} = \underline{\hspace{2cm}}$ Current gain $A_i = i_3 / i_1 = \underline{\hspace{2cm}}$ Power gain $A_p = A_v A_i = \underline{\hspace{2cm}}$ Amplifier? Yes ☐ No ☐

11. In the I/V converter shown in the figure, $R_1 = 100 \text{ k}\Omega$. Find its trans-resistance and input resistance.



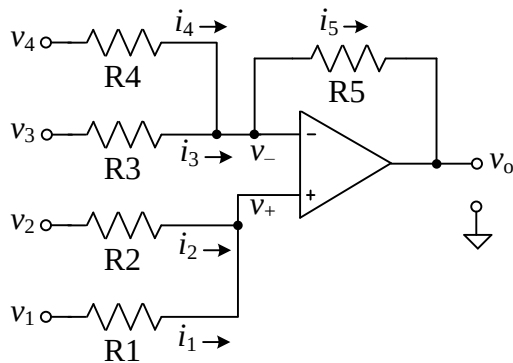
Marks: 2 (= 1 + 1)

Answer:

$$R = v_o / i_{in} = \underline{\hspace{2cm}}$$

$$R_{in} = v_{in} / i_{in} = \underline{\hspace{2cm}}$$

12. In the summer circuit shown in the figure, $R_1 = R_2 = R_3 = 10 \text{ k}\Omega$, $R_4 = R_5 = 100 \text{ k}\Omega$. Find the voltage gain and input resistance for the first and third inputs.



Marks: 4 (= 1 x 4)

Answer: (show steps)

$$A_{v1} = v_o / v_1 = \underline{\hspace{2cm}}$$

$$R_{in1} = v_1 / i_1 = \underline{\hspace{2cm}}$$

$$A_{v3} = v_o / v_3 = \underline{\hspace{2cm}}$$

$$R_{in3} = v_3 / i_3 = \underline{\hspace{2cm}}$$